Practical – 2

Aim of the Practical :

To configure a Local Area Network (LAN) with multiple switches and PCs, assign static IP addresses, and verify connectivity using Packet Tracer simulation.

Software Required:

1. Cisco Packet Tracer
2. 2 Switches (2960)
3. 4 PCs
4. Copper straight-through cables

Procedure:

1. Topology Design
    - Drag and drop 4 PCs and 2 switches into the Packet Tracer workspace.
    - Connect:
        - PC-A and PC-B → Switch0
        - PC-C and PC-D → Switch1
        - Switch0 ↔ Switch1 (using a copper straight-through cable).

2. Assign IP Addresses (static)
    - Configure the following addresses:(Since all PCs are in the same LAN, a gateway is not required.)
        - PC-A - 192.168.1.1
        - PC-B - 192.168.1.2
        - PC-C - 192.168.1.3
        - PC-D - 192.168.1.4

3. Verify Connectivity
    - On PC-A, open Command Prompt → type ping 192.168.1.2. ○ Similarly, ping PC-C and PC-D (ping 192.168.1.3, ping 192.168.1.4).
    - Successful replies confirm LAN connectivity.

4. Expected Output:
    - All PCs should be able to ping each other within the LAN.
    - No packet loss should occur.

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Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=7ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 1ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time=7ms TTL=128
Reply from 192.168.1.4: bytes=32 time=7ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 3ms

C:\>192.168.1.2
Invalid Command.

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
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